MAEG4060 Virtual Reality Systems and Applications

Course Project — User Guide

Interactive Physical Simulation of Off-Road Vehicle Dynamics

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About this guide. This PDF explains how to **launch** the DirectX 9 sandbox, what **hardware** is expected, how **keyboard and gamepad** inputs map to the vehicle, and how to fix common **runtime issues**. Build-from-source and CI notes are grouped at the end for developers.

1 Launch the application

1.1 Game build (markers and end users)

- Open the `Game_Build` folder from your submission package (or use a folder produced by `cmake -install` / `scripts/package_game_build.bat`).
- **Double-click** `maeg4060_stage2.exe`. Keep the executable in the same directory as the bundled `Assets` tree.

Tip. If meshes or textures fail to load, you are almost certainly running the `.exe` with the wrong working directory. Launch from the folder that contains both `maeg4060_stage2.exe` and `Assets/`, or confirm that relative paths such as `Assets/models/...` resolve. Optional replacement OBJ/MTL props live under `Assets/models/replacements/` (see `README.md` there).

2 Hardware and software

- **Platform:** 64-bit **Windows** (recommended).
- **Graphics: DirectX 9.0c** rendering path; GPU and drivers should support D3D9. Install the **DirectX 9 End-User Runtime** (or ensure equivalent system components) if you see missing-D3D errors.
- **Display:** At least **1280×720**; the window can be resized.
- **Input (optional): XInput-compatible** gamepad (e.g. Xbox controller) for analog throttle and steering. Keyboard control works without a pad.

Figure 1: *Example window: terrain splats, vehicle, props, windmill, and HUD.*

3 Controls

3.1 Keyboard

Key	Action
W / S	Forward / reverse (throttle)
A / D	Steer left / right
Space	Brake (or hand brake, depending on build)
Esc	Exit the application

Bindings follow the project implementation (`GetAsyncKeyState`). If your package includes on-screen hints, treat those as authoritative for that build.

3.2 Gamepad (XInput)

Control	Action
Left stick	Steering
Right trigger	Throttle (accelerate)
Left trigger	Brake / partial reverse

Triggers approximate analog throttle; keyboard W/S remain digital.

3.3 Windmill behaviour (default)

The windmill **plays automatically by default**. If the shipped windmill asset contains keyframed animation data, the hierarchy animation is advanced each frame; otherwise the blades use a simple code-driven spin for the static mesh path.

4 Troubleshooting

- **Missing DLLs / DirectX errors:** Repair or install the DirectX 9 end-user runtime; update the GPU driver.
- **Windows protected your PC (SmartScreen):** This is expected for an unsigned student executable. Click **More info** then **Run anyway**.
- **Black screen or device creation failure:** Prefer windowed operation first, close other exclusive full-screen apps, then retry after a driver update.
- **Assets not found:** Run from the folder layout in Section 1; verify `Assets/` sits beside the executable.
- **Gamepad ignored:** Prefer a wired connection or official wireless adapter; confirm Windows lists the device as XInput.

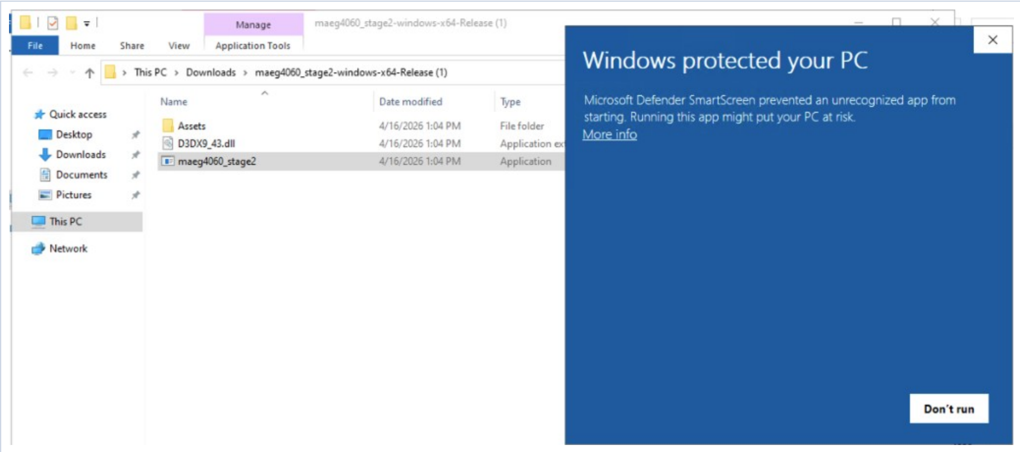

Figure 2: *Microsoft Defender SmartScreen: click* ***More info*** *to reveal the override option.*

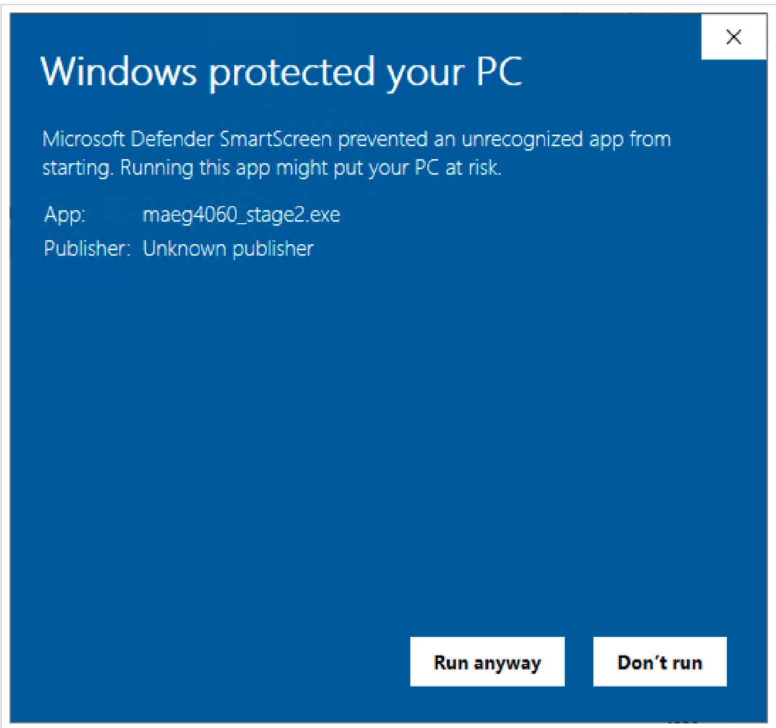

Figure 3: *Microsoft Defender SmartScreen: click* ***Run anyway*** *to start* `maeg4060_stage2.exe`.

5 Build from source (developers)

The CMake target maeg4060_stage2 expects legacy **D3DX** headers and import libraries (d3dx9.h, d3dx9math.h, . . .), which the Windows 10/11 SDK alone does not provide. Configuration uses either:

- **June 2010 DirectX SDK:** set DXSDK_DIR to an installation whose Include/ and Lib/ trees CMake can consume; *or*
- **NuGet D3DX:** leave DXSDK_DIR unset and allow CMake to fetch Microsoft.DXSDK.D3DX on first configure (requires network; first configure may take several minutes).

Suggested command line (matches scripts/configure_build_release.bat):

Run vcvars64.bat from your Visual Studio install, then from the repository root:

```
cmake -B build -G Ninja -DCMAKE_BUILD_TYPE=Release ^
  -DCMAKE_CXX_COMPILER=cl -DCMAKE_C_COMPILER=cl
cmake --build build
```

Ninja with an explicit -DCMAKE_CXX_COMPILER=cl avoids some generator detection issues seen with the Visual Studio CMake workflow. If you switch generators, delete the build directory (or wipe CMakeCache.txt and CMakeFiles/) before reconfiguring.

If the compiler reports **C1060** (compiler out of heap space), rebuild with less parallelism, e.g. cmake -build build -j 1.

Alternative (Visual Studio generator):

```
cmake -B build -G "Visual Studio 17 2022" -A x64
cmake --build build --config Release
```

Use this when the toolset is detected correctly.

Package output: run scripts/package_game_build.bat, or cmake -install build -prefix Game_Build, to populate Game_Build/.

6 GitHub Actions (optional)

If enabled on your fork, .github/workflows/windows-build.yml builds on windows-latest and publishes a zip containing maeg4060_stage2.exe, D3DX9_43.dll when the NuGet D3DX path is used, and a copy of Assets/. Unzip on Windows and run the executable from that folder with Assets alongside it.